

Formulas*discrete time growth:*

- $N_T = N_0 \lambda^T$
- $\lambda = f + p$
- $\mathcal{R} = f/(1 - p)$

continuous time growth:

- $N(t) = N(0) \exp(rt)$
- $r = b - d$
- $\mathcal{R} = b/d$

structured growth:

- $\ell_x = p_1 \times p_2 \times \dots p_{x-1}$
- $\mathcal{R} = \sum \ell_x f_x$

competition:

- α_{ij} = effect **of** species i **on** species j
- $C = \alpha_{12}\alpha_{21}$
- $E_{ij} = \alpha_{ij}K_i/K_j$

1. We believe that in a particular desert, Joshua trees and cactuses are competing only for water. According to our simple theory, we would expect:

- A. A tendency for co-existence
- B. The species with the larger value of K (indiv/ha) to dominate
- C. The species with the larger value of α (measuring its impact on the other species) to dominate
- D. The species with the larger value of E (measuring its impact on the other species) to dominate
- E. A tendency for mutual exclusion

2. Which of the following is *not* a possible advantage of producing fewer offspring with the same total amount of energy (thus investing more in each offspring)?

- A. Greater ability to help offspring survive
- B. Greater ability to help offspring disperse
- C. Greater ability to produce high-quality offspring
- D. More energy left over to produce offspring in future years

3. A scientist has made a life table for a perennial population of deer, with censusing *before* reproduction. If she re-does the life table with a census after reproduction, she would expect $\mathcal{R}$ to be _____ because population dynamics look _____ when viewed before reproduction.

- A. higher; fastest
- B. higher; slowest
- C. lower; fastest
- D. lower; slowest
- E. the same; the same

4. Which of the following traits is *not* typically associated with r strategies?

- A. Fast life cycle
- B. Efficient resource use
- C. Relatively small investment per offspring
- D. Relatively large investment in dispersal

Use the following information for the next two questions. A population of hemlock trees is estimated to be at stable age distribution, with a constant life table, with reproductive number $\mathcal{R} = 1.2$. It takes the trees several decades to reach maturity and reproduce.

5. This population is

- A. declining
- B. stable
- C. increasing
- D. showing damped oscillations
- E. there is not enough information to answer this question

6. What is the *most accurate* statement you can make about the finite rate of growth λ , measured with a time step of one year?

- A. We expect $\lambda > 1.2$
- B. We expect $\lambda = 1.2$
- C. We expect $\lambda > 1$
- D. We expect $1.2 > \lambda > 1$
- E. We expect $1.2 > \lambda$

7. A species of small plant is highly adapted to dramatic changes in conditions (ranging from very good growth conditions to very bad). We would expect it to:

- A. Have a very fast life cycle
- B. Have a very slow life cycle
- C. Have adaptations that speed up its life cycle in good conditions and slow it down in bad conditions
- D. Have adaptations that speed up its life cycle in bad conditions and slow it down in good conditions

8. We expect populations to be regulated (kept under control). This typically works because their _____ tends to decrease when the population _____.

- A. growth rate; has been established for a long time
- B. growth rate; becomes larger
- C. projected future size; has been established for a long time
- D. projected future size; becomes larger

9. Japanese maples are trying to invade a stable Ontario forest dominated by sugar maples (we don't know whether they will be successful). Based on these assumptions, we would expect the reproductive number of the _____ population to satisfy _____.

- A. sugar maple; $\mathcal{R} > 1$
- B. Japanese maple; $\mathcal{R} > 1$
- C. sugar maple; $\mathcal{R} = 1$
- D. Japanese maple; $\mathcal{R} = 1$

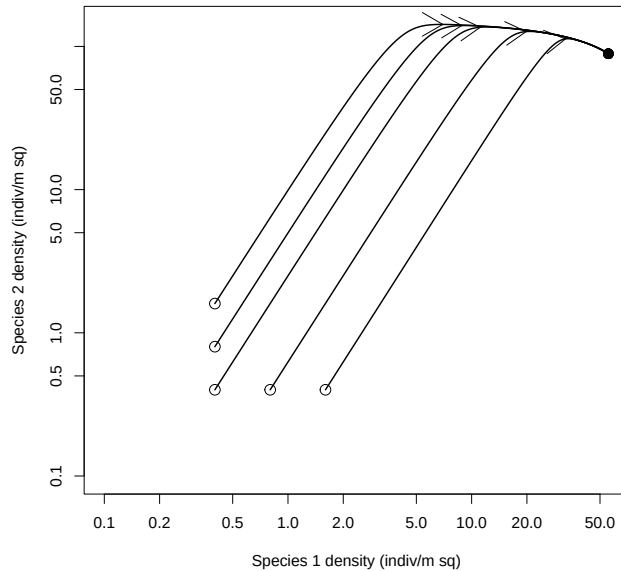
10. In an environment where seed survival is highly variable from year to year, we expect that semelparous plants (reproducing only once) are likely to invest relatively _____ than iteroparous plants in _____.

- A. more; mechanisms for seed dispersal
- B. less; mechanisms for seed dispersal
- C. more; each individual seed
- D. less; each individual seed

11. Which of these is *not* an effect that tends to *balance* the tendency of populations to grow or decline exponentially without limit?

- A. Allee effects
- B. Competition for access to mates
- C. Competition for access to resources
- D. Resource depletion
- E. Dynamics of natural enemies

Use the picture above for the next three questions. The path in the middle corresponds to a starting density of 0.4 indiv/m^2 for each species.



12. This picture shows what sort of competition outcome?

- A. Species 1 dominates
- B. Species 2 dominates
- C. The species co-exist
- D. There is mutual exclusion

13. Assuming density dependence can be neglected at low densities, which species has the higher value of $r_{\max}$?

- A. Species 1
- B. Species 2
- C. They have the same value of $r_{\max}$
- D. Either species can have a higher value, depending on initial conditions.

14. From this picture, we can conclude that:

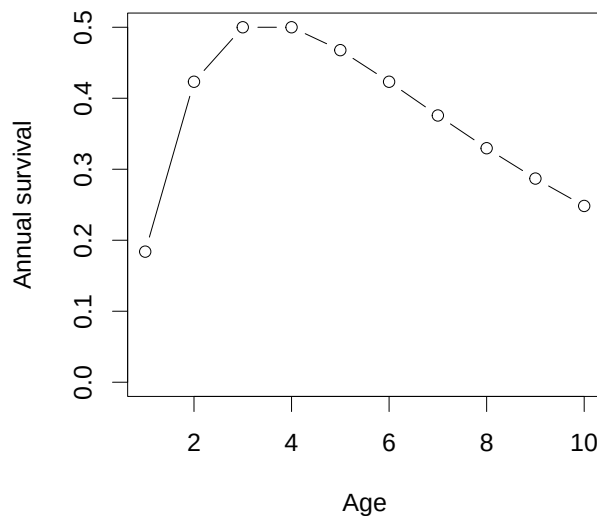
- A. Both of the competition coefficients $\alpha < 1$.
- B. Both of the competition coefficients $\alpha > 1$.
- C. Both of the population-level competitive effects $E < 1$.
- D. Both of the population-level competitive effects $E > 1$.
- E. We cannot make any of these conclusions from this picture.

15. A simple population model has *structure*, but not *regulation* (individuals are assumed to be independent). Cohorts are not independent. If the model has $\mathcal{R}_0 > 1$, then: The modeled population _____ grow exponentially at first, and _____ grow exponentially as it approaches a stable age distribution (SAD)

- A. will; will
- B. may not; will
- C. will; may not
- D. may not; may not

16. The “balance argument” for sexual allocation implies that, if individuals do not interact strongly with their siblings, females should on average:

- A. Produce the same number of male and female offspring
- B. Use the same amount of resources *per offspring* for male and female offspring
- C. Use the same *total* amount of resources for male and female offspring
- D. All of the above

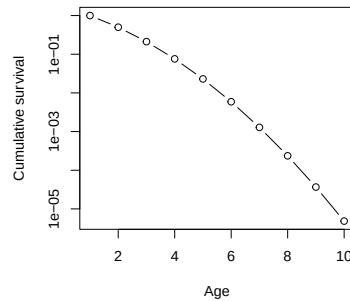
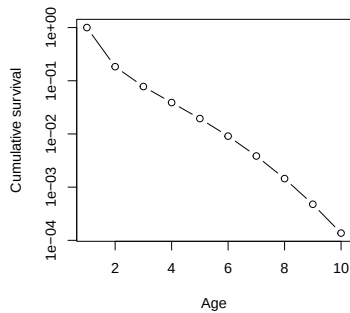
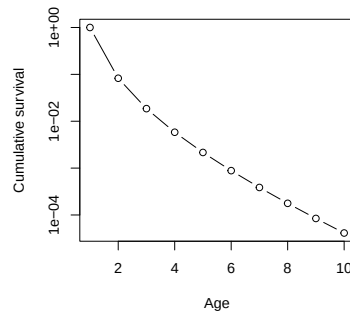
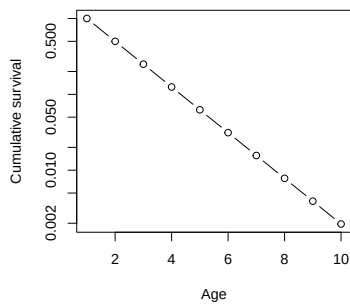


Use the picture above for the following 2 questions.

17. What does this picture of survivorship in an idealized age-structured population indicate about *mortality* in this population?

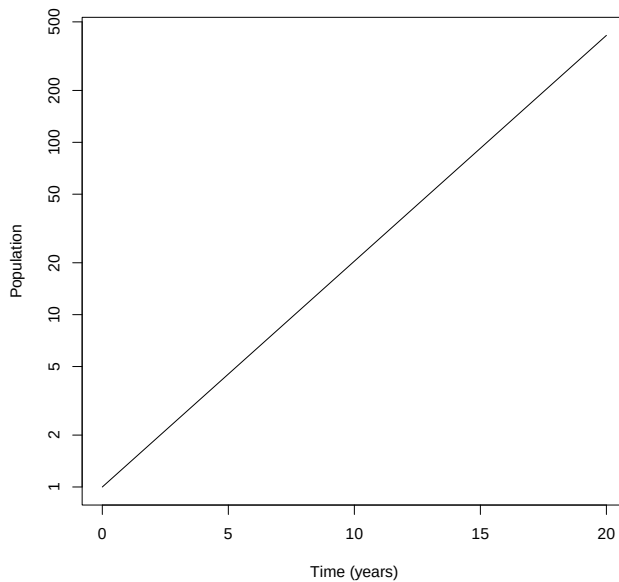
- A. Mortality is constant
- B. Mortality is elevated in older individuals
- C. Mortality is elevated in younger individuals
- D. Mortality is elevated in both older and younger individuals

18. The pictures below show *cumulative* survival. Which one corresponds to the picture shown above?

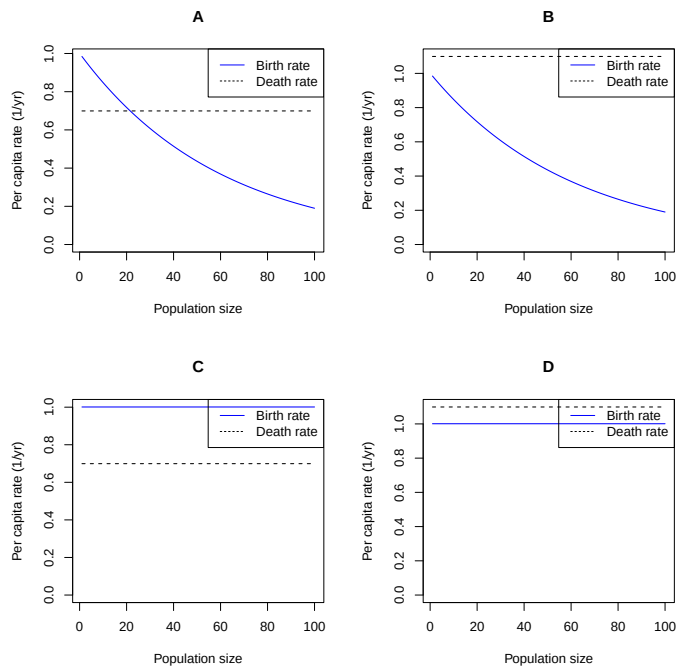


- A. Upper left
- B. Upper right
- C. Lower left
- D. Lower right

Use the picture below for the next two questions. It shows a time series for a continuous-time birth-death model.



19. This picture shows a population that is:
- A. Increasing arithmetically
 - B. Increasing geometrically
 - C. Increasing arithmetically on the log scale, but geometrically on a linear scale
 - D. Increasing geometrically on the log scale, but arithmetically on a linear scale
20. Which of the small pictures shows the assumptions that generated the time plot?



- A. A
- B. B
- C. C
- D. D

Short-answer questions

Answer questions *in pen*. *Briefly* show necessary work and equations. Points may be *deducted* for wrong information, even when the correct information is also there.

21. An annual plant has variable fecundity: each female has an average of 2 successful female offspring in an good year (half of the time) and an average of 0.4 successful female offspring in a bad year (the other half of the time).

a) (1 point) What is the long-run average λ for this species?

b) (1 point) What are two possible strategies this plant could evolve to spread out the risk of a bad year (hedge its bets)?

22. A species of falcon and a species of hawk are living in the same region and competing to eat songbirds and small mammals that they find there.

a) What is a possible reason why the competition between species might be relatively *stronger* than within species in this case?

b) What is a possible reason why the competition between species might be relatively *weaker* than within species in this case?

c) What are *two* possible outcomes of the competitive interaction in the second case (when between-species competition is weak)?